



- i Fire duct to be tested / assessed to BS 476: Part 24 for all sizes up to 25 meters x 3 meters cross-sectional area and fully certified to vertical and horizontal plane.
- j Fire rated duct fabricated to Method 3 of BS 5588: Part 9, factory produced. The coating compound shall be applied either offsite or onsite on the ground, dried and cured.
- k Fire duct expansion under fire conditions shall not exceed following,
 - at 430⁰C an expansion of 0.006106mm per mm
 - at 600⁰C an expansion of 0.00852mm per mm
 - At 1100⁰C an expansion of 0.01562mm per mm.

19) **INSULATION**

19.1. GENERAL:

Scope of this specification comprises of supplying, installing, testing and commissioning of insulation on sheet metal ducts, pumps, chilled water piping, condensate drain piping, refrigerant piping, air handling equipments, chillers, expansion tank, AHU room and duct acoustic lining.

- In addition to complying with the relevant standards, all insulating material shall be free from Asbestos.

Comply

- All insulation finishes and coverings shall be classified as Class 1 (one) surface spread when tested in accordance with B.S. 476, Part 7.

Comply

- All adhesives, mastics, coatings, sealers and primers shall be classified as Class 1 Surface spread when tested in accordance with B.S. 476, part 7. They shall not in any way attack the insulation or the surface to which the insulation is being applied and shall be suitable for the working temperatures.

Noted

- All material delivered to site shall be new and fully dried out and so maintained throughout the progress of the works. All insulating materials shall be stored in storage sheds, and in accordance with the Manufacturer's recommendations.

Noted

- In order to ensure that the insulation applied is in all respects in accordance with the specification, sections shall, as required by the Consultant, be cut from the finished insulation. The Contractor is to allow in his price for the removal and replacement of two sections of each type of insulation. If, however, defects are revealed, further sections shall be cut out for inspection, and all cutout sections shall be replaced at no cost to the CLIENT.



- If further defects are revealed then the Consultant shall have the right, when in his opinion it is necessary, to issue instructions for any part or the whole of the insulation to be removed and replaced. The replacement with new insulation shall be to the satisfaction of the Consultant and the cutting out and replacing shall be at no cost to the Contract.
- Particular attention shall be paid to the finished appearance of all thermal insulation which must present a neat and symmetrical appearance running true in line with pipe layouts, etc.
- Any rough, irregular and badly finished surfaces shall be stripped down and reinsulated to the Consultant's satisfaction.
- In certain cases the type of insulation specified will require a painted finish, in addition to identifying bands and any signs which are to be applied.
- All systems are to have been tested and approved by the Consultant prior to installation of insulation.
- In case of heavy ducts and pipes wherein the supports are fixed directly on ducts / pipes the supports shall be also insulated with a thermal break / barrier.

19.2. Duct Insulation

Comply

a) Insulation material shall be Closed Cell Elastomeric Nitrile Rubber with 7 mils factory laminated glass cloth on one side.

Comply

b) Thermal conductivity of insulation material shall not exceed 0.035 W/ (m.K) at mean temperature of 0°C as per EN 12667.

Comply

c) Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 7,000 without any external barrier as per EN 12086.

Comply

d) The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.

Comply

e) The insulation material shall have fire performance of V0, HB as per UL 94, 1996.

Comply

f) The insulation material shall be FM (Factory Mutual), USA Approved.

g) Density of insulation material shall be between 40 to 60 Kg /m³.

Comply

h) The insulation material shall be dust and fiber free.

i) The insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of 0 Deg.C as per EN 14706.

Comply

j) Glass Cloth: Weight shall be 205 +/- 10 grams per square meter, Tensile Strength shall be 275 +/- 25 Kg / 50 mm (minimum) and Thickness shall be 0.18 mm / 7 mil. Glass Cloth shall be UV resistant for artificial weathering.

Comply

k) The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.

l) Thickness of the insulation shall be as specified for the individual application.

m) The insulation material shall be installed as per manufacturer's recommendation.



Installation

External thermal insulation shall be provided as follows:

The thickness of insulation material shall be as per design/manufacture standard. Following procedure shall be adhered to:

Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90° cut in thickness of sheet. Hacksaw or blades are not acceptable tools for cutting the insulation.

Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Self Adhesive aluminum tapes / silver tapes as per the insulation material finish as per manufacturer recommendations. The insulation installers shall be certified by manufacturer.

Direct contact between Duct and hanger shall be avoided. Hangers shall pass outside the Duct Support. Insulation Manufacturer shall supply duct supports having PET foam as load taking member embedded in flexible elastomeric foam nitrile rubber sheet with outer metal face having aluminium metal) so that the insulation material is joint with insulation material on both side of pipe support for Nitrile and the weight of pipe is transferred to the PET saddle.

Noted

19.3. PIPING INSULATION

Scope:

Supply and fix thermal insulation for piping, ducting and equipment as per the specification described in the section.

Materials of Insulation:

The following insulating materials shall be used:

a) For Piping

GENERAL

All underground and aboveground chilled water lines shall be insulated with pre-insulated PUF.

INSULATION

The service pipe insulation shall be polyurethane foam with 36 kg/cu m minimum density, 90% minimum closed cell content, minimum compressive strength of 40 psi and thermal conductivity of 0.14 Btu-in/hr/ft²/OF. The insulation shall completely fill the annular space between the service pipe and jacket and shall be bonded to both, the service pipe & jacket.

Comply